The original embedded OS. It was first developed in 1982 by SCG. Currently being supported by NXP Semiconductor through TriMedia VLIW core which they brought from Wind River Systems. pSOS supports priority inheritance and priority ceiling. It also supports segmented memory management. An example application of pSOS is a base station in a cellular network.

vxWorks

Supported Platform: x86, x86-64, MIPS, PowerPC, SH-4, ARM, SPARC Version 8 (V8)

This operating system has been to Mars, twice. It was used in both Pathfinder and Curiosity mission. It comes with it's own Eclipse-based IDE called Tornado. Development is carried out using the host-target paradigm the developers can use the superior facilities of developing on a 'host' system like an editor, a compiler toolchain, a debugger, and an emulator. The software is then compiled to the 'target' system.

VRTX

Supported Platform: ARM, MIPS, PowerPC, RISC

Kernel Size (for VRTXmc): 4K to 8K ROM and 1K RAM

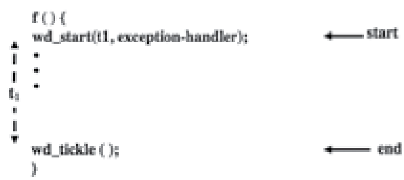
VRTX is developed by Mentor Graphics. It comes in two versions: VRTXmc (micro-controller) for small systems requiring minimal memory use and VRTXsa (scalable architecture) for full operating system features. VRTX powers the Hubble Space Telescope it has also been by the US FAA (Federal Aviation Authority) for use in avionics. VRTXmc was used extensively by Motorola in many of it's mobile phones.

Windows CE

Supported Platform: x86, MIPS, ARM, SuperH

Kernel Size: 350K

Windows CE currently billed as Windows Embedded Compact is the embedded offering from the PC giant. Microsoft directly licenses Windows CE to OEM and manufacturers they can modify the user interface while Windows CE provides the technological backbone. It provides 256 priority levels. To optimise performance, all the threads run in the kernel mode.



Use of one shot timers in functions with critical section